

Deep Learning

Deep learning is a subfield of machine learning based on artificial neural networks containing multiple layers.

A neural network normally contains an input layer, hidden layers, and an output layer. The network learns parameters from training data.

Convolutional neural networks are widely used for computer vision. Transformers are widely used for language and other sequential tasks.

Training involves forward propagation, calculating a loss function, backpropagation, and optimization.

Deep learning is used for image classification, object detection, speech recognition, natural language processing, medical imaging, recommendation systems, and cybersecurity.